

Project Design Phase-II

Data Flow Diagram & User Stories

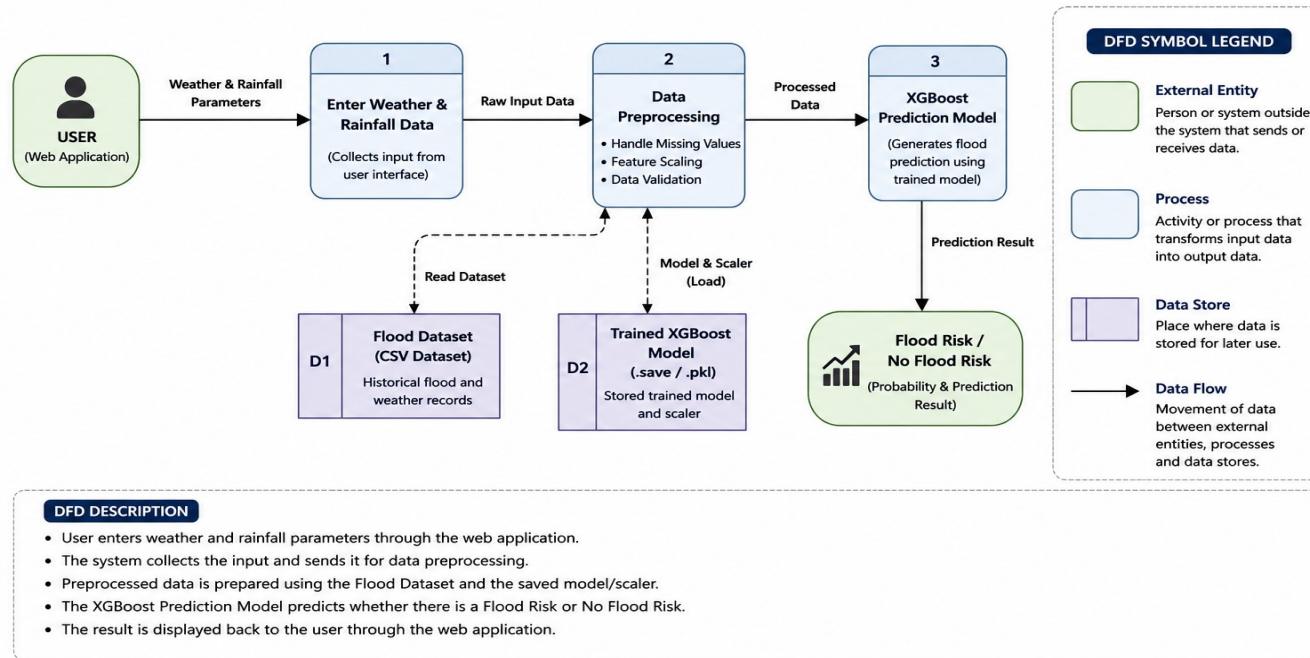
Date	17 July 2026
Team ID	-----
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

DATA FLOW DIAGRAM (LEVEL-0)

Project : Rising Waters – A Machine Learning Approach to Flood Prediction



Data Flow Diagram (Level 0)

External Entity: User

Process: Flood Prediction System

Data Store: Flood Dataset & Trained XGBoost Model

Flow: User enters weather/rainfall data → Flask Application → ML Model → Prediction Result (Flood / No Flood) → User.

User Stories

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Administrator	Dataset Management	USN-1	As an administrator, I want to upload the historical flood dataset so that the system can use it for model training.	Dataset is uploaded successfully and available for preprocessing.	High	Sprint-1
Data Analyst	Data Exploration	USN-2	As a data analyst, I want to visualize and analyze rainfall and flood data so that I can identify important patterns and relationships.	Charts, heatmaps, and summary statistics are generated correctly.	High	Sprint-1
Data Scientist	Data Preprocessing	USN-3	As a data scientist, I want to clean the dataset by handling missing values, outliers, and feature scaling so that the data is suitable for machine learning.	Clean and preprocessed dataset is generated successfully.	High	Sprint-1
Data Scientist	Feature Engineering	USN-4	As a data scientist, I want to split the dataset into training and testing sets so that the model can be trained and evaluated accurately.	Training and testing datasets are created successfully.	High	Sprint-1
ML Engineer	Model Training	USN-5	As a machine learning engineer, I want to train Decision Tree, Random Forest, KNN, and XGBoost models to identify the best-performing algorithm.	All models are trained and evaluation metrics are generated.	High	Sprint-2
ML Engineer	Model Evaluation	USN-6	As a machine learning engineer, I want to compare model performance using accuracy, confusion matrix, precision, recall, and F1-score so that I can select the best model.	XGBoost is selected based on the highest performance metrics.	High	Sprint-2

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Administrator	Model Management	USN-7	As an administrator, I want to save the trained XGBoost model and preprocessing scaler so that they can be used during deployment.	Model and scaler are stored successfully and loaded without errors.	High	Sprint-2
User (Resident)	Flood Prediction	USN-8	As a resident, I want to enter rainfall and weather parameters into the web application so that I can know the flood risk in my area.	User inputs are validated and prediction is generated successfully.	High	Sprint-3
User (Resident)	Prediction Result	USN-9	As a resident, I want to receive a clear prediction indicating Flood Risk or No Flood Risk so that I can take preventive measures.	Prediction result is displayed accurately within a few seconds.	High	Sprint-3
Disaster Management Officer	Decision Support	USN-10	As a disaster management officer, I want to use prediction results to issue early warnings and plan emergency response activities.	Reliable prediction results are available to support decision-making.	High	Sprint-3
Administrator	Web Application Deployment	USN-11	As an administrator, I want to deploy the trained model using a Flask web application so that users can access the prediction system online.	Web application loads successfully and provides real-time predictions.	Medium	Sprint-4
Administrator	Model Maintenance	USN-12	As an administrator, I want to periodically update the dataset and retrain the model so that prediction accuracy remains high with new data.	Updated model is retrained, tested, and redeployed successfully		